

PAJU, Korea, Republic of

WMO: 470990

Lat: 37.8858N

Lon: 126.7664E

Elev: 101

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	7.2	-9.4	3.3	13.0	-5.7	4.1	14.7	13.0	25.3	11.7	25.2	2.0	140	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	89.5	77.5	87.1	75.6	84.8	73.9	79.8	86.1	78.5	83.9	77.2	82.2	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.2	146.8	83.1	77.0	141.0	81.8	75.7	134.9	80.3	43.6	86.7	42.2	83.9	40.9	82.6	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.3	10.9	9.5	-2.1	93.3	6.0	1.5	-6.4	94.4	-9.9	95.3	-13.3	96.2	-17.6	97.3		
			-2.9	81.2	5.7	2.2	-7.0	82.8	-10.3	84.2	-13.5	85.4	-17.7	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	23.8	29.5	40.0	51.5	62.8	71.2	76.4	77.7	68.8	56.5	42.5	27.9	(6)
(7)		DBStd	19.66	8.13	7.79	6.67	6.04	4.79	3.74	3.68	4.55	4.59	6.62	8.28	8.14	(7)
(8)		HDD50	2712	812	574	318	52	0	0	0	0	21	250	685		(8)
(9)		HDD65	5664	1277	993	775	404	100	2	0	17	270	676	1150		(9)
(10)		CDD50	3628	0	0	7	99	397	637	818	858	565	222	25	0	(10)
(11)		CDD65	1109	0	0	0	1	32	190	353	394	132	7	0	0	(11)
(12)		CDH74	7857	0	0	0	23	316	1182	2393	3248	661	34	0	0	(12)
(13)		CDH80	2129	0	0	0	3	41	262	645	1092	85	1	0	0	(13)
(14)	Wind	WSAvg	3.5	3.0	3.7	4.4	4.6	4.0	3.4	3.5	3.4	3.2	3.0	3.3	3.1	(14)
(15)	Precipitation	PrecAvg	55.6	0.6	1.1	1.7	3.7	3.9	5.3	18.1	10.6	5.8	2.0	1.9	0.9	(15)
(16)		PrecMax	84.0	2.0	2.8	3.6	6.9	6.4	14.9	43.4	23.9	18.3	7.2	4.3	2.5	(16)
(17)		PrecMin	27.8	0.1	0.0	0.5	1.5	0.7	1.7	7.0	1.5	1.2	0.4	0.3	0.2	(17)
(18)		PrecStd	14.0	0.5	0.8	1.0	1.3	1.7	3.3	9.4	7.0	4.8	1.7	1.1	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	55.5	65.5	77.0	84.0	88.7	91.2	93.4	84.9	77.8	67.5	50.7	(19)
(20)		MCWB	39.4	46.0	50.7	57.8	62.7	70.1	77.6	79.8	71.6	64.1	57.5	43.5		(20)
(21)		2%	DB	41.7	49.9	60.0	71.5	80.6	85.3	87.9	90.9	82.4	74.0	62.7	46.7	(21)
(22)		MCWB	35.2	40.6	48.2	54.4	62.3	68.4	76.3	78.9	69.8	61.7	54.8	41.5		(22)
(23)		5%	DB	38.9	45.7	56.0	67.8	77.4	83.0	85.9	88.7	80.4	71.0	59.2	43.3	(23)
(24)		MCWB	33.3	38.9	45.7	52.8	61.2	68.3	75.4	77.8	68.6	60.1	51.9	38.4		(24)
(25)		10%	DB	36.3	41.9	52.5	64.2	74.7	80.7	83.8	86.1	78.0	67.9	55.8	40.1	(25)
(26)		MCWB	31.8	35.6	43.5	51.6	60.2	67.7	74.7	76.9	67.5	58.0	49.8	35.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.1	49.5	54.4	62.3	68.0	74.5	80.6	82.5	75.7	67.9	60.5	46.6	(27)
(28)		MDCB	45.0	53.4	60.8	73.5	75.3	81.4	87.6	89.2	80.1	73.2	63.9	49.2		(28)
(29)		2%	WB	36.6	42.2	50.9	58.1	65.7	72.8	78.9	80.7	73.6	64.5	57.0	42.1	(29)
(30)		MDCB	40.2	48.9	57.1	64.5	73.7	79.7	84.5	87.6	78.3	70.3	60.6	46.1		(30)
(31)		5%	WB	34.4	39.1	47.6	55.6	64.0	71.4	77.7	79.4	71.7	62.3	53.5	38.8	(31)
(32)		MDCB	38.0	44.7	54.1	63.3	72.5	77.9	82.5	85.4	76.1	68.2	58.1	42.6		(32)
(33)		10%	WB	31.9	36.4	44.5	53.3	62.3	70.0	76.6	78.3	70.1	60.0	50.5	36.3	(33)
(34)		MDCB	35.9	41.2	51.0	61.4	71.2	76.6	80.9	83.7	75.2	65.9	54.9	39.8		(34)
(35)	Mean Daily Temperature Range		MDBR	19.8	19.8	20.5	21.9	21.2	17.2	12.2	14.0	17.4	21.9	19.0	17.5	(35)
(36)		5% DB	MCDBR	20.2	23.5	25.1	27.7	26.6	21.8	16.4	16.9	19.6	23.0	21.7	19.0	(36)
(37)			MCWBR	14.5	15.5	14.5	12.9	10.7	7.7	6.5	6.5	8.3	10.9	13.5	14.1	(37)
(38)			MCDBR	17.5	21.2	21.1	21.8	20.1	15.5	12.2	14.4	14.6	19.3	18.3	17.9	(38)
(39)		5% WB	MCWBR	13.5	15.3	14.1	12.1	9.6	6.8	5.7	6.4	7.6	10.5	13.4	14.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.425	0.485	0.564	0.616	0.627	0.700	0.672	0.578	0.508	0.503	0.481	0.426	(40)	
(41)		taud	1.934	1.837	1.712	1.650	1.676	1.596	1.704	1.920	2.029	1.917	1.913	1.962	(41)	
(42)		Ebn at Noon	223	226	221	220	221	205	209	227	235	218	202	211	(42)	
(43)		Edh at Noon	44	56	69	78	77	84	75	59	50	50	45	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	793	1034	1351	1574	1769	1655	1286	1378	1322	1133	766	691	(44)	
(45)		RadStd	50	78	128	112	171	174	193	189	140	77	76	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	+187	+103

Nomenclature: See separate page